

Algorithm Design VIII

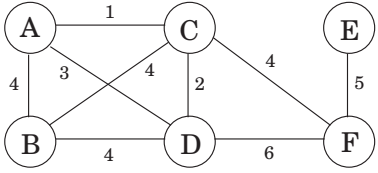
Greedy Algorithms

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Minimum Spanning Trees

Build a Network

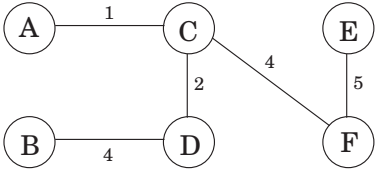


Suppose you are asked to network a collection of computers by linking selected pairs of them.

This translates into a graph problem in which

- nodes are computers,
- undirected edges are potential links, each with a maintenance cost.

Build a Network



The goal is to

- pick enough of these edges that the nodes are connected,
- the total maintenance cost is minimum.

One immediate observation is that the optimal set of edges cannot contain a cycle.

Properties of the Optimal Solutions

Lemma (1)

Removing a cycle edge cannot disconnect a graph.

So the solution must be connected and acyclic: undirected graphs of this kind are called trees.

A tree with minimum total weight, is a minimum spanning tree, MST.

Input: An undirected graph $G = (V, E)$; edge weights w_e

Output: A tree $T = (V, E')$ with $E' \subseteq E$ that minimizes

$$\texttt{weight}(T) = \sum_{e \in E'} w_e$$

Lemma (2)

A tree on n nodes has $n - 1$ edges.

To build the tree one edge at a time, starting from an empty graph.

Each of the n nodes is disconnected from the others, in a connected component by itself.

As edges are added, these components merge. Since each edge unites two different components, exactly $n - 1$ edges are added by the time the tree is fully formed.

When a particular edge (u, v) comes up, we can be sure that u and v lie in separate connected components, for otherwise there would already be a path between them and this edge would create a cycle.

Lemma (3)

Any connected, undirected graph $G = (V, E)$ with $|E| = |V| - 1$ is a tree.

It is the converse of Lemma (2). We just need to show that G is acyclic.

While the graph contains a cycle, remove one edge from this cycle.

The process terminates with some graph $G' = (V, E'), E' \subseteq E$, which is acyclic and, by Lemma (1), is also connected.

Therefore G' is a tree, whereupon $|E'| = |V| - 1$ by Lemma (2). So $E' = E$, no edges were removed, and G was acyclic to start with.

Lemma (4)

An undirected graph is a tree if and only if there is a unique path between any pair of nodes.

In a tree, any two nodes can only have one path between them; for if there were two paths, the union of these paths would contain a cycle.

On the other hand, if a graph has a path between any two nodes, then it is connected. If these paths are unique, then the graph is also acyclic.

A Greedy Approach

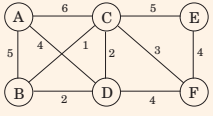
Kruskal's minimum spanning tree algorithm starts with the empty graph and then selects edges from E according to the following rule.

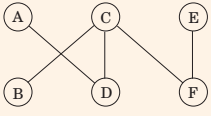
Repeatedly add the next lightest edge that doesn't produce a cycle.

Example

Starting with an empty graph and then attempt to add edges in increasing order of weight

$$B-C; C-D; B-D; C-F; D-F; E-F; A-D; A-B; C-E; A-C$$





The Cut Property

Lemma

Suppose edges X are part of a MST of $G = (V, E)$. Pick any subset of nodes S for which X does not cross between S and $V \setminus S$, and let e be the lightest edge across this partition. Then

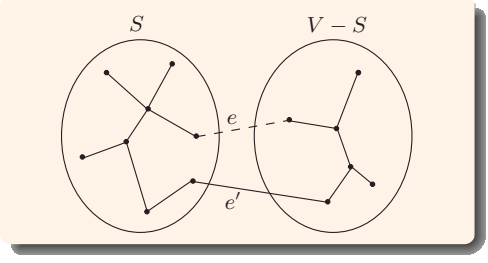
$$X \cup \{e\}$$

is part of some MST.

The Cut Property

A cut is any partition of the vertices into two groups, S and $V \backslash S$.

It is safe to add the lightest edge across any cut, provided X has no edges across the cut.



Proof of the Cut Property

Proof:

Edges X are part of some MST T; if the new edge e also happens to be part of T, then there is nothing to prove.

So assume e is not in T. We will construct a different MST T' containing $X \cup \{e\}$ by altering T slightly, changing just one of its edges.

Add edge e to T. Since T is connected, it already has a path between the endpoints of e, so adding e creates a cycle.

This cycle must also have some other edge e' across the cut $(S, V \backslash S)$. If we now remove e'

$$T' = T \cup \{e\} \backslash \{e'\}$$

which we will show to be a tree.

T' is connected by Lemma (1), since e' is a cycle edge. And it has the same number of edges as T; so by Lemma (2) and Lemma (3), it is also a tree.

Proof of the Cut Property

Proof:

T' is a minimum spanning tree, since

$$weight(T') = weight(T) + w(e) - w(e')$$

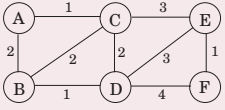
Both e and e' cross between S and $V \backslash S$, and e is the lightest edge of this type. Therefore $w(e) \leq w(e')$, and

$$weight(T') \leq weight(T)$$

Since T is an MST, it must be the case that $weight(T') = weight(T)$ and that T' is also an MST.

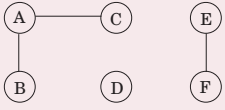
An Example of Cut Property

(a)

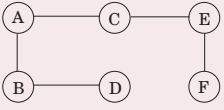


(b)

Edges X:

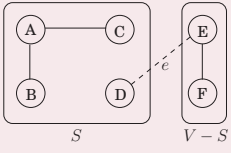


MST T:

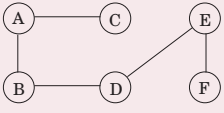


(c)

The cut:



MST T':



Kruskal's Algorithm

`KRUSKAL` (G, w)

input : A connected undirected graph $G = (V, E)$, with edge weight w_e

output: A minimum spanning tree defined by the edges X

for *all* $u \in V$ **do**
    `makeset` (u);
end
$X = \{\}$;
Sort the edges E by weight;
for *all* $(u, v) \in E$ *in increasing order of weight* **do**
    if `find` $(u) \neq$ `find` (v) **then**
        add (u, v) to X;
        `union` (u,v)
    end
end

Data Structure Retailer: Disjoint Sets

`makeset`(x)	create a singleton set containing x	$\lvert V\rvert$
`find`(x)	find the set that x belong to	$2 \cdot \lvert E\rvert$
`union`(x, y)	merge the sets containing x and y	$\lvert V\rvert - 1$

Union by rank

We store a set is by a directed tree. Nodes of the tree are elements of the set, arranged in no particular order, and each has parent pointers that eventually lead up to the root of the tree.

This root element is a convenient representative, or name, for the set. It is distinguished from the other elements by the fact that its parent pointer is a self-loop.

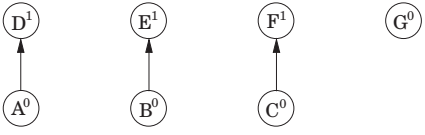
In addition to a parent pointer π, each node also has a rank that, for the time being, should be interpreted as the height of the subtree hanging from that node.

An Example

After $\texttt{makeset}(A), \texttt{makeset}(B), \ldots, \texttt{makeset}(G)$:

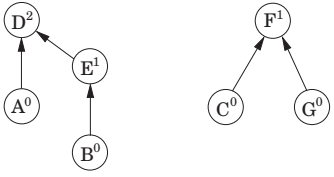


After $\texttt{union}(A, D), \texttt{union}(B, E), \texttt{union}(C, F)$:

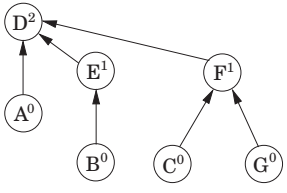


An Example

After $\texttt{union}(C, G), \texttt{union}(E, A)$:



After $\texttt{union}(B, G)$:



Prim's Algorithm

A General Kruskal's Algorithm

$X = \{\}$;
repeat until $|X| = |V| - 1$;
    pick a set $S \subset V$ for which X has no edges between S and $V - S$;
    let $e \in E$ be the minimum-weight edge between S and $V - S$;
    $X = X \cup \{e\}$;

Prim's Algorithm

A popular alternative to Kruskal's algorithm is Prim's, in which the intermediate set of edges X always forms a subtree, and S is chosen to be the set of this tree's vertices.

On each iteration, the subtree defined by X grows by one edge.

The lightest edge between a vertex in S and a vertex outside S. We can equivalently think of S as growing to include the vertex $v \notin S$ of smallest `cost`:

$$\texttt{cost}(v) = \min_{u \in S} w(u, v)$$

The Algorithm

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PRIM(G, w)
input : A connected undirected graph $G = (V, E)$, with edge weights w_e
output: A minimum spanning tree defined by the array $prev$

for *all* $u \in V$ **do**
    $cost(u) = \infty$;
    $prev(u) = nil$;
end
pick any initial node u_0;
$cost(u_0) = 0$;
H =makequeue (V) \\ *using cost-values as keys*;
while H *is not empty* **do**
    v=deletemin (H);
    for *each* $(v, z) \in E$ **do**
        if $cost(z) > w(v, z)$ **then**
            $cost(v) = w(v, z)$; $prev(z) = v$;
            decreasekey (H,z);
        end
    end
end

Dijkstra's Algorithm

DIJKSTRA(G, l, s)
input : Graph $G = (V, E)$, directed or undirected; positive edge length $\{l_e \mid e \in E\}$; Vertex $s \in V$
output: For all vertices u reachable from s, $dist(u)$ is the set to the distance from s to u

for *all* $u \in V$ **do**
    $dist(u) = \infty$;
    $prev(u) = nil$;
end
$dist(s) = 0$;
H =makequeue(V) \\ *using dist-values as keys*;
while H *is not empty* **do**
    u=deletemin(H);
    for *all edge* $(u, v) \in E$ **do**
        if $dist(v) > dist(u) + l(u, v)$ **then**
            $dist(v) = dist(u) + l(u, v)$; $prev(v) = u$;
            decreasekey (H,v);
        end
    end
end

Think About

Let C be a cycle with no red edges, and select an uncolored edge of C of max cost and color it red.

Let D be a edge set crossing a cut with no blue edges, and select an uncolored edge in D of min cost and color it blue.

Apply the red and blue rules nondeterministically until all edges are colored.

The blue edges form an MST.

Set Cover

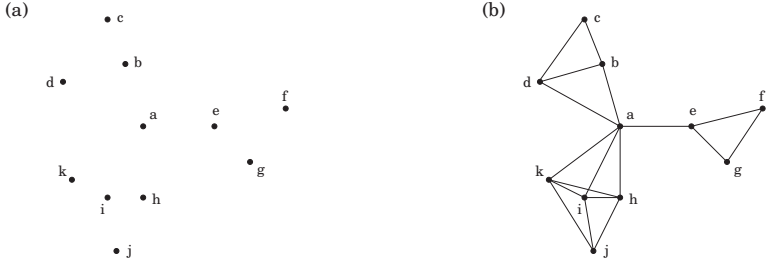
The Problem

A county is in its early stages of planning and is deciding where to put schools.

There are only two constraints:

- each school should be in a town,
- and no one should have to travel more than 30 miles to reach one of them.

Q: What is the minimum number of schools needed?



The Problem

This is a typical (cardinality) set cover problem.

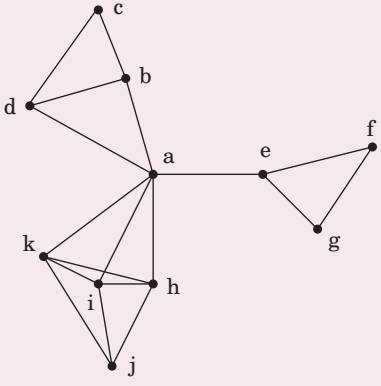
- For each town x, let S_x be the set of towns within 30 miles of it.
- A school at x will essentially "cover" these other towns.
- The question is then, how many sets S_x must be picked in order to cover all the towns in the county?

Set Cover Problem

Set Cover

- Input: A set of elements B, sets $S_1, \ldots, S_m \subseteq B$
- Output: A selection of the S_i whose union is B.
- Cost: Number of sets picked.

The Example



Performance Ratio

Lemma

Suppose B contains n elements and that the optimal cover consists of OPT sets. Then the greedy algorithm will use at most $\ln n \cdot OPT$ sets.

Proof.

Let n_t be the number of elements still not covered after t iterations of the greedy algorithm (so $n_0 = n$).

Since these remaining elements are covered by the optimal OPT sets, there must be some set with at least n_t/OPT of them.

Therefore, the greedy strategy will ensure that

$$n_{t+1} \leq n_t - \frac{n_t}{OPT} = n_t(1 - \frac{1}{OPT})$$

which by repeated application implies

$$n_t \leq n_0(1 - \frac{1}{OPT})^t$$

A more convenient bound can be obtained from the useful inequality

$$1 - x \leq e^{-x} \text{ for all } x$$

with equality if and only if $x = 0$,

Thus

$$n_t \leq n_0(1 - \frac{1}{OPT})^t < n_0(e^{-\frac{1}{OPT}})^t = ne^{-\frac{t}{OPT}}$$

At $t = \ln n \cdot OPT$, therefore, n_t is strictly less than $ne^{-\ln n} = 1$, which means no elements remain to be covered.

Why Greedy Does Not Work: Coin Changing

Coin changing

Goal. Given U. S. currency denominations $\{1, 5, 10, 25, 100\}$, devise a method to pay amount to customer using fewest coins.

Example $\$34$.

Cashier's algorithm. At each iteration, add coin of the largest value that does not take us past the amount to be paid.

Example $\$2.89$.

Cashier's algorithm

CASHIERS-ALGORITHM($x, c_1, c_2, \ldots, c_n$)

SORT n coin denominations so that $0 < c_1 < c_2 < \ldots < c_n$;
$S \leftarrow \emptyset$;
while $x > 0$ **do**
    $k \leftarrow$ largest coin denomination c_k such that $c_k \leq x$;
    if *no such* k **then** RETURN no solution;
    else
        $x \leftarrow x - c_k$;
        $S \leftarrow S \cup \{k\}$;
    end
end
RETURN S;

Is the cashier's algorithm optimal?

A Yes, greedy algorithms are always optimal.

B Yes, for any set of coin denominations $c_1 < c_2 < \ldots < c_n$ provided $c_1 = 1$.

C Yes, because of special properties of U.S. coin denominations.

D No.

Cashier's algorithm (for arbitrary coin denominations)

Q. Is cashier's algorithm optimal for any set of denominations?

A. No. Consider U.S. postage: $1, 10, 21, 34, 70, 100, 350, 1225, 1500$.

- Cashier's algorithm: $\$140 = 100 + 34 + 1 + 1 + 1 + 1 + 1$.
- Optimal: $\$140 = 70 + 70$.

A. No. It may not even lead to a feasible solution if $c_1 > 1$: $7, 8, 9$.

- Cashier's algorithm: $\$15 = 9 + ?$.
- Optimal: $\$15 = 7 + 8$.

Properties of any optimal solution (for U.S. coin denominations)

Property. Number of pennies ≤ 4.

Proof. Replace 5 pennies with 1 nickel.

Property. Number of nickels ≤ 1.

Property. Number of quarters ≤ 3.

Property. Number of nickels + number of dimes ≤ 2.

Proof.

- Recall: ≤ 1 nickel.
- Replace 3 dimes and 0 nickels with 1 quarter and 1 nickel;
- Replace 2 dimes and 1 nickel with 1 quarter.

Optimality of cashier's algorithm (for U.S. coin denominations)

Theorem

Cashier's algorithm is optimal for U.S. coins $\{1, 5, 10, 25, 100\}$.

A rather formal proof

Proof. by induction on amount to be paid x

Consider optimal way to change $c_k \leq x \leq c_{k+1}$: greedy takes coin k.

Claim that any optimal solution must take coin k.

- if not, it needs enough coins of type $c_1, \ldots, c_{k-1}$ to add up to x.
- table below indicates no optimal solution can do this

Problem reduces to coin-changing $x - c_k$ cents, which, by induction, is optimally solved by cashier's algorithm.

k	c_k	all optimal solutions must satisfy	max value of $c_1, c_2, \ldots c_{k-1}$ in any optimal solution
1	1	$P \leq 4$	none
2	5	$N \leq 1$	4
3	10	$N + D \leq 2$	$4 + 5 = 9$
4	25	$Q \leq 3$	$20 + 4 = 24$
5	100	no limit	$75 + 24 = 99$